NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 6: L7-L9

Question 1 (Lecture 7)

What is the relationship between Machine Learning (ML) and Artificial Intelligence (AI)?

Question 2 (Lecture 7)

Suppose you are working on a machine learning algorithm to navigate a robot through a room from start to goal while avoiding obstacles within some time, according to previous runs through the room with sensor feedback (e.g., successful and failed attempts). What are the Task, Performance, and Experience involved according to the definition of machine learning?

Question 3 (Lecture 7)

Suppose you are working on a machine learning algorithm that groups customers into market segments based on purchasing behavior.

(i) What type of task is it?

(a) Regression

(b) Classification

(c) Clustering

(d) None of these

(ii) If the data you have collected involve millions of attributes, what would you do?

Question 4 (Lecture 7)

You are given a set of data for supervised learning. A sample block of data looks like this:

$$\begin{matrix} 0.0012 & 20123 & 1.545 & 12.47 & 1 \\ 0.0034 & 29087 & 1.908 & 20.66 & -1 \\ 0.0029 & 19456 & 2.109 & 27.09 & 1 \\ 0.0013 & 32098 & 1.009 & 18.76 & 1 \\ 0.0088 & 25908 & 1.398 & 30.08 & -1 \end{matrix}$$

Each row corresponds to a sample data measurement with 4 input features and 1 response.

(i) What kind of undesired effect can you anticipate if this set of raw data is used for learning?

(ii) How can the data be preprocessed to handle this issue?

Question 5 (Lecture 7)

Please download the synthetic data “synthetic-data.xlsx” provided in the folder “EE2213/Files/Tutorials” on Canvas. You may need to install openpyxl library to read .xlsx files. Read the dataset by “from pandas import read_excel”. Use Python to perform the following tasks.

(i) Print the summary statistics of this dataset. (You may use “.describe()” or “.info()”)

(ii) Check and locate if any missing values or duplicates. (You may use ".isnull()" and ".duplicate()")

Question 6 (Lecture 8)

Given the following dataset for training:

Sample No.	x	y
1	1.6	3.1
2	1.5	3.4
3	1.0	2.9
4	1.1	3.0
5	1.2	3.8
6	0.9	2.9

(i) Perform a linear regression, sketch the result of line fitting, calculate the mean squared error for the training set, and predict a test point {x=0.5}. Hint: you might want to utilize "from sklearn.metrics import mean_squared_error" for calculating MSE, and "import matplotlib.pyplot as plt" for plotting.

(ii) Perform a 3rd-order polynomial regression, sketch the result of line fitting, calculate the mean squared error for the training set, and predict the same test point {x=0.5}. Hint: you might want to utilize "from sklearn.preprocessing import PolynomialFeatures" to generate polynomial features.

(iii) Does it have a 6th-order polynomial regression solution? If so, proceed to solved it. Otherwise, apply ridge regression with different regularization parameter λ:[0.0001, 0.01, 1].

Question 7 (Lecture 8)

The California Housing Dataset is used for predicting the median house value for a given district in California, based on features from the 1990 U.S. Census. Get the dataset by "from sklearn.datasets import fetch_california_housing" and load the dataset as DataFrame by "fetch_california_housing(as_frame=True)". Use Python to perform the following tasks.

(i) Split the database into three sets: 60% of samples for training, 20% of samples for validation, and 20% of samples for testing. Hint: you might want to utilize "from sklearn.model_selection import train_test_split" for the splitting.

(ii) Use Pearson's correlation as a feature selection metric to select the top 2 features that have the highest absolute correlations with the target output. Use the selected top 2 features to construct the training, validation and test datasets. Hint: you might want to utilize ".corr()" for calculating Pearson's correlation coefficients.

(iii) Apply the polynomial model from orders 1 to 6 to the dataset generated from part (ii) without L2 regularization unless $\mathbf{P}^T\mathbf{P}$ is not invertible (then, use $\lambda = 0.00001$). Plot Mean Squared Errors (MSE) over polynomial orders for both training and the validation sets. Which polynomial order provides the best MSE in the training and validation sets? Which polynomial order should be selected? Use the selected polynomial order to evaluate the

MSE for the test set. Hint: you might want to utilize "from sklearn.metrics import mean_squared_error" for calculating MSE.

(iv) Use regularization parameter λ =1 for all polynomial orders and repeat the same analyses. Compare the plots of (iii) and (iv). What do you see?

Question 8 (Lecture 9)

The Breast Cancer Wisconsin Diagnostic Dataset is used for classifying tumors as either malignant (labelled 0) or benign (labelled 1), based on digitized measurements of breast mass cell nuclei. Get the dataset by "from sklearn.datasets import load_breast_cancer" and load the dataset as DataFrame by "load_breast_cancer (as_frame=True)". Use Python to perform the following tasks.

(i) Split the database into three sets: 80% of samples for training, 10% of samples for validation, and 10% of samples for testing. Hint: you might want to utilize "from sklearn.model_selection import train_test_split" for the splitting.

(ii) Use Pearson's correlation as a feature selection metrics. First, select features that have absolute correlations greater than 0.5. Second, among the selected features, remove features that have absolute correlations with each other greater than 0.9. Use the selected features to construct the training, validation and test datasets. Hint: you might want to utilize ".corr()" for calculating Pearson's correlation coefficients.

(iii) Apply linear regression for classification to the dataset generated in part (ii) without regularization unless $\mathbf{X}^T\mathbf{X}$ is not invertible (then, use $\lambda = 0.00001$). The threshold is set to 0.5 for class prediction. Compute the training, validation and test accuracies. Demonstrate the confusion matrix for the test set. Hint: you might want to utilize "from sklearn.metrics import accuracy_score, confusion_matrix" for evaluation metrics.

(iv) Apply logistic regression to the dataset generated from part (ii) without regularization. The threshold is set to 0.5 for class prediction. The number of iterations for training is set to 20000. Try different learning rates: 0.1, 0.01 and 0.001. Plot the cost over training iterations for each learning rate. Which learning rate provides the best accuracy in the training and the validation sets? Which learning rate should be selected? Use the selected learning rate to compute the accuracy and confusion matrix for the test set.

Question 9 (Lecture 9)

The Iris dataset is used to classify 3 species of iris flower: Setosa, Versicolor, and Virginica. Get the dataset by "from skelarn.datasets import load_iris". Use Python to perform the following tasks.

(i) Split the database into three sets: 80% of samples for training, 10% of samples for validation, and 10% of samples for testing. Hint: you might want to utilize "from sklearn.model_selection import train_test_split" for the splitting.

(ii) Construct the target output using one-hot encoding. Hint: you might want to utilize "from sklearn.preprocessing import OneHotEncoder".

(iii) Apply linear regression for classification to the dataset generated in part (ii) without regularization unless $\mathbf{X}^T\mathbf{X}$ is not invertible (then, use $\lambda = 0.00001$). Compute the training, validation and test accuracies. Demonstrate the confusion matrix for the test set. Hint: you might want to utilize "from sklearn.metrics import accuracy_score, confusion_matrix" for evaluation metrics.

(iv) Perform logistic regression for classification with different learning rates: 0.1, 0.01 and 0.001. The number of iterations for training is set to 20000. Plot the cost over training iterations for each learning rate. Which learning rate provides the best accuracy in the training and the validation sets? Which learning rate should be selected? Use the selected learning rate to compute the accuracy and confusion matrix for the test set.

Question 10 (Lecture 9)

Tabulate the confusion matrix for the following classification problem. The class-1, class-2, class-3, and class-4 data points are indicated by circles, diamonds, triangles, and squares.